

Homework Assignment 5

Questions:

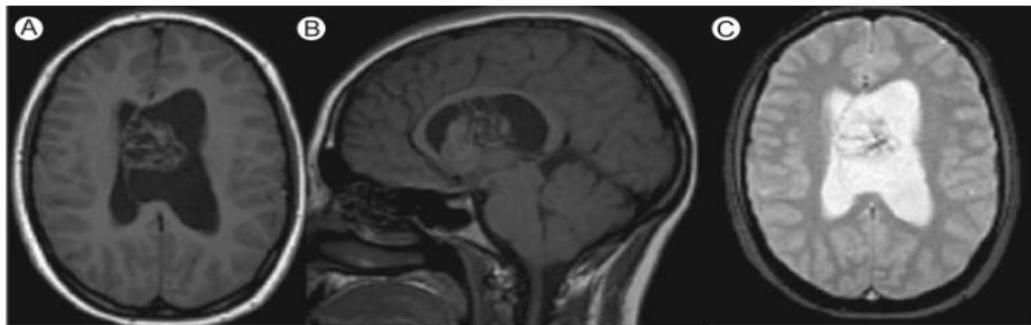
1. What is the brain parenchyma and what does it comprise of?

When analyzing the visual images that result from a CT scan allows us to differentiate between three different structures: the skull (appears white), the brain parenchyma (appears grey), and the air within sinuses (appears black). The “brain parenchyma” is more specifically comprised of the following (all appearing grey): white matter, gray matter, muscle, and hemorrhage (blood which appears a very light grey).

2. What is a cerebral edema and do brain hemorrhage and or hydrocephalus cause them?

When fluid (Blood in the case of hemorrhage, or CSF in the case of hydrocephalus) fills Intracranial space in abnormal places (places it shouldn't be) and can lead to the “Mass Effect” where other structures get displaced or compressed.

3. Below are two T1-weighted MRI scans and a single T2-weighted MRI scan. This patient has a central neurocytoma, a solid mass attached to the septum pellucidum.



- a. After comparing these images to T1 and T2-weighted images in the lecture slides and taking note of the patient's additional symptoms of nausea, headache, and cognitive troubles, slides, what is the condition this patient is experiencing? Detail the type of condition (there are two different types of this condition) and why you think it is this type.

I believe this patient is suffering from “noncommunicating hydrocephalus” after comparing these T1 and T2 weighted MRI scans to those in the lecture slides. In doing so, we see the T1-weighted image for this patient compared to their T2-weighted image clearly visualizes their abnormally enlarged ventricles. The change in color from black (in T1) to white (in T2) indicates that this fluid is CSF and not blood. Therefore, that make this patient's condition one of hydrocephalus, and I believe that it is more specifically of the “noncommunicating” type because we can see from the second T1-weighted image (figure B) that the CSF-buildup/mass-effect is localized to structures deeper within the brain (and not the arachnoid granulations).

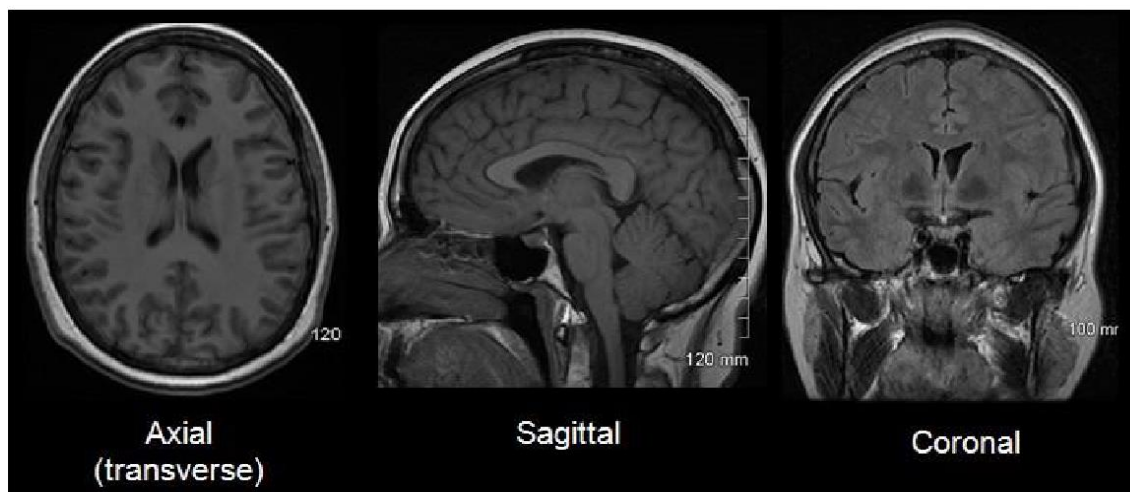
- b. Using your best judgement, is herniation occurring in this case study?

No, the patient is not experiencing herniation in this case because they do not portray any of the typical symptoms: dilated pupils, hemiparesis, respiratory arrest, blood pressure instability, or death. In fact, my diagnosis of hydrocephalus in “part a” of this question is further supported by the patient’s reported symptoms: nausea, headache, and cognitive troubles. This buildup of CSF is likely due to a “CSF flow obstruction” at/near the septum pellucidum (as seen in the scans).

4. Lawrence is experiencing some car trouble on La Salle and is parked on the shoulder searching for the owner’s manual inside his car. Meanwhile, Benny is driving his motorcycle down La Salle and realizes he is running late to a biker rally. As he picks up speed, he fails to notice Lawrence opening his car door. Benny collides with the door, is flung from his motorcycle, and hits his head on the pavement. The doctors at the hospital say he has fluid in his brain as they try to stabilize his blood pressure and control his breathing. Judging from Benny’s symptoms, **what portion of his brain was likely damaged in the accident? Name the associated condition.**

Lawrence is likely experiencing a “central herniation,” a downward displacement of his brainstem. The damage to areas of his brainstem and surrounding areas, in direct consequence of the pressure caused by blood build-up, would provide an appropriate explanation as for his instable blood pressure and loss in respiratory (breath) control he faces in the hospital.

5. Draw representations of the three primary imaging planes.



Matching:

Match the following terms to their definitions. Highlight the terms that are part of Cushing's Triad.

A. Papilledema	4. Engorgement or elevation of the optic disc that leads to blurred vision or loss of sight
B. Hypertension	5. Increased blood pressure
C. Diplopia	1. Double vision
D. Meningitis	2. Bacterial infection that causes inflammation of the meninges; symptoms include stiff neck, headache and fever
E. Bradycardia	3. Slow heart rate of < 60 BPM
F. Mass Effect	7. Distortion of the typical brain anatomy through displacement and compression of structures due to increased mass
G. Primary Brain Tumor	6. Tumor that originates in the brain
H. Metastatic Brain Tumor	8. Tumor starts in another part of the body and then spreads to the brain

1. Double vision
2. Bacterial infection that causes inflammation of the meninges; symptoms include stiff neck, headache and fever
3. Slow heart rate of < 60 BPM
4. Engorgement or elevation of the optic disc that leads to blurred vision or loss of sight
5. Increased blood pressure
6. Tumor that originates in the brain
7. Distortion of the typical brain anatomy through displacement and compression of structures due to increased mass
8. Tumor starts in another part of the body and then spreads to the brain

Below are CT Scans of different hematomas. Using the lecture slides and other resources. Define each type of hematoma and match each to a scan. Your options are **epidural**, **subarachnoid**, **intracerebral**, and **subdural**.



This CT scan represents an “epidural hematoma:” bleeding between dura and skull, caused by trauma (mostly due to injury of temporal bone).

- We can also see the characteristic “lens shape” of this hematoma



This CT scan represents a “subarachnoid hematoma:” bleeding in CSF-filled space between the arachnoid and pia.

- You can see on this scan how far the pia extends into the cortex due to the pressure of an intracranial bleed pushing on it as well as the arachnoid mater



This CT scan represents a “subdural hematoma:” bleeding between dura and arachnoid, caused by rupture of the bridging veins.

- We can also see the characteristic “crescent shape” of this hematoma



This CT scan represents an “intracerebral hematoma:” a bleed inside of the brain itself

- You can see on this scan fresh blood which appears to be accumulating in the center of patient’s left hemisphere near the central sulcus and extending laterally outwards towards the cranium